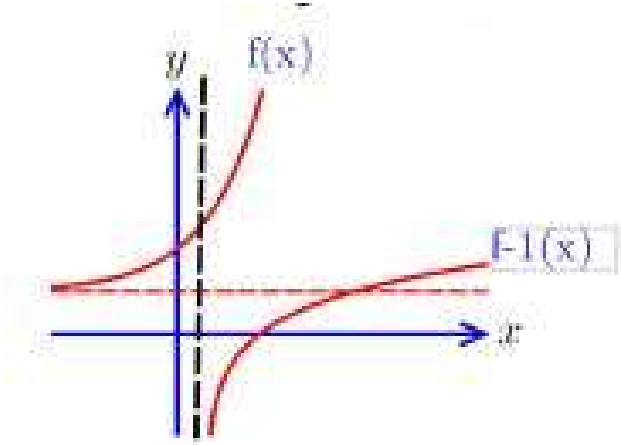
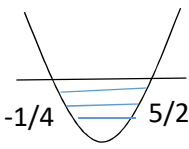


NO	SCHEME	MARKS
1.	$f(x) = 1 + \ln x$ $ff^{-1}(x) = x$ $1 + \ln(f^{-1}(x)) = x$ $\ln(f^{-1}(x)) = x - 1$ $f^{-1}(x) = e^{x-1}$ $R_{f^{-1}} = [1, \infty)$ $f^{-1}(3) = e^{3-1} = e^2$	 K1 K1 J1 B1 K1J1
2.	$\log_9 x = (\log_3 x)^2$ $\frac{\log_3 x}{\log_3 9} = (\log_3 x)^2$ $\frac{\log_3 x}{\log_3 3^2} = (\log_3 x)^2$ $\frac{\log_3 x}{2} = (\log_3 x)^2$ $\log_3 x = 2(\log_3 x)^2$ $u = \log_3 x$ $u = 2u^2$ $2u^2 - u = 0$ $u(2u - 1) = 0$ $u = 0 \quad u = \frac{1}{2}$ $\log_3 x = 0 \quad \log_3 x = \frac{1}{2}$ $x = 3^0 \quad x = 3^{\frac{1}{2}}$ $x = 1 \quad x = \sqrt{3}$ $x > 1, \therefore x = \sqrt{3}$	 K1 K1 K1 J1 K1 J1 J1

3.	$2^{2x-2} - 2^{x+1} = 2^x - 2^3$ $2^{2x}2^{-2} - 2^x2^1 = 2^x - 8$ $\frac{(2^x)^2}{4} - 2(2^x) - 2^x + 8 = 0$ $(2^x)^2 - 12(2^x) + 32 = 0$ $u = 2^x$ $u^2 - 12u + 32 = 0$ $(u - 8)(u - 4) = 0$ $u - 8 = 0 \quad u - 4 = 0$ $u = 8 \quad u = 4$ $2^x = 8 \quad 2^x = 4$ $x = 3 \quad x = 2$	K1 K1 K1 K1 J1 K1 J1
4. a	$g(x) = \frac{kx+8}{4x-5}$ $g \circ g(x) = x$ $\frac{k\left(\frac{kx+8}{4x-5}\right) + 8}{4\left(\frac{kx+8}{4x-5}\right) - 5} = x$ $\frac{k\left(\frac{kx+8}{4x-5}\right) + 8\left(\frac{4x-5}{4x-5}\right)}{4\left(\frac{kx+8}{4x-5}\right) - 5\left(\frac{4x-5}{4x-5}\right)} = x$ $\frac{k^2x + 8k + 32x - 40}{4kx + 32 - 20x + 25} = x$ $\frac{k^2x + 8k + 32x - 40}{4kx + 32 - 20x + 25} = x$ $k^2x + 8k + 32x - 40 = x(4kx + 32 - 20x + 25)$ $k^2x + 8k + 32x - 40 = 4kx^2 + 32x - 20x^2 + 25x$ $(k^2 + 32)x + 8k - 40 = (4k - 20)x^2 + 57x$ compare $4k - 20 = 0$ $k = 5$	K1 K1 K1 K1 J1

4b	$g(x_1) = g(x_1)$ $\frac{kx_1 + 8}{4x_1 - 5} = \frac{kx_2 + 8}{4x_2 - 5}$ $(kx_1 + 8)(4x_2 - 5) = (kx_2 + 8)(4x_1 - 5)$ $4kx_1x_2 - 5kx_1 + 32x_2 - 40 = 4kx_1x_2 - 5kx_2 + 32x_1 - 40$ $-5kx_1 + 32x_2 = -5kx_2 + 32x_1$ $-5kx_1 + 32x_2 + 5kx_2 - 32x_1 = 0$ $-x_1(5k + 32) + x_2(32 + 5k) = 0$ $(x_2 - x_1)(5k + 32) = 0$ $x_2 \neq x_1 \quad 5k + 32 = 0$ $k = -\frac{32}{5}$	K1 K1 K1 K1 J1
5 a	$ff^{-1}(x) = x$ $e^{3f^{-1}(x)} + 4 = x$ $e^{3f^{-1}(x)} = x - 4$ $3f^{-1}(x) = \ln(x - 4)$ $f^{-1}(x) = \frac{\ln(x - 4)}{3}$	K1 K1 K1 J1
b	 $D_f = (-\infty, \infty) \quad D_{f^{-1}} = (4, \infty)$	R1 R1 R1 R1 B1 B1

6a	$\frac{p}{4-2i} + \frac{q}{4+2i} = 1 + \frac{5}{2}i$ $\frac{p(4+2i) + q(4-2i)}{(4-2i)(4+2i)} = 1 + \frac{5}{2}i$ $\frac{4p + 2pi + 4q - 2qi}{16 - 4i^2} = 1 + \frac{5}{2}i$ $\frac{4p + 4q + (2p - 2q)i}{20} = 1 + \frac{5}{2}i$ $\frac{4p + 4q}{20} = 1 \quad \frac{2p - 2q}{20} = \frac{5}{2}$ $p + q = 5 \quad p - q = 25$ $p = 15, \quad q = -10$	K1 K1 K1 K1 J1J1
b	$\log_{10} 2 = m \quad \log_{10} 7 = n$ $(14^{3x+1})(8^{2x+3}) = 7$ $(2 \times 7)^{3x+1} (2^3)^{2x+3} = 7$ $2^{3x+1+6x+9} \times 7^{3x+1} = 7$ $2^{9x+10} \times 7^{3x+1} = 7$ $\log(2^{9x+10} \times 7^{3x+1}) = \log 7$ $\log 2^{9x+10} + \log 7^{3x+1} = \log 7$ $(9x+10)\log 2 + (3x+1)\log 7 = \log 7$ $(9x+10)m + (3x+1)n = n$ $9xm + 10m + 3xn + n = n$ $9xm + 3xn = -10m$ $x(9m + 3n) = -10m$ $x = \frac{-10m}{9m + 3n}$	K1 K1 K1 K1 K1 K1 J1

7a	$ 2-3x < x+3 $	K1
	$\sqrt{(2-3x)^2} < \sqrt{(x+3)^2}$	K1
	$(2-3x)^2 < (x+3)^2$	K1
	$4-12x+9x^2 < x^2+6x+9$	K1
	$3x^2-18x-5 < 0$	K1
	$(2x-5)(4x+1) < 0$	K1
	$x = \frac{5}{2} \quad x = -\frac{1}{4}$	
		K1J1
	$\left\{x : -\frac{1}{4} < x < \frac{5}{2}\right\}$	J1
bi	$2x-1 = 2x+2-3 = 2(x+1)-3$	K1
	$x+1 < 0$	K1
	$\Rightarrow 2(x+1)-3 < 0$	J1
	$\therefore 2x-1 < 0$	
	$\frac{2x-1}{x+1} > 2$	
	$\frac{2x-1}{x+1} - 2 > 0$	
	$\frac{2x-1}{x+1} - 2 = \frac{2x-1-2(x+1)}{x+1}$	K1
	$= \frac{2x-1-2x-2}{x+1} = \frac{-3}{x+1}$	J1
	$x+1 < 0$	
	$\frac{-3}{x+1} > 0$	K1
	$\frac{2x-1}{x+1} - 2 > 0$	
	$\frac{2x-1}{x+1} > 2$	J1

8.a	$\lim_{x \rightarrow 0} \frac{mx + 3x^2}{4x - 8x^2} = 3$ $\lim_{x \rightarrow 0} \frac{x(m + 3x)}{x(4 - 8x)} = 3$ $\lim_{x \rightarrow 0} \frac{(m + 3x)}{(4 - 8x)} = 3$ $\frac{(m + 3(0))}{(4 - 8(0))} = 3$ $\frac{m}{4} = 3$ $m = 12$	K1 K1 J1
b	$\lim_{x \rightarrow 0} \frac{\sqrt{3-x} - \sqrt{3}}{x} = \lim_{x \rightarrow 0} \frac{\sqrt{3-x} - \sqrt{3}}{x} \times \frac{\sqrt{3-x} + \sqrt{3}}{\sqrt{3-x} + \sqrt{3}}$ $= \lim_{x \rightarrow 0} \frac{3 - x - 3}{x(\sqrt{3-x} + \sqrt{3})}$ $= \lim_{x \rightarrow 0} \frac{-x}{x(\sqrt{3-x} + \sqrt{3})}$ $= \lim_{x \rightarrow 0} \frac{-1}{\sqrt{3-x} + \sqrt{3}}$ $= \frac{-1}{\sqrt{3-0} + \sqrt{3}} = -\frac{1}{2\sqrt{3}}$	K1 K1 K1J1
9.a	$y = \cos ec \left\{ \sin \left\{ \ln(x+1) \right\} \right\}$ $\frac{dy}{dx} = -\cos ec \left\{ \sin \left\{ \ln(x+1) \right\} \right\} \cot \left\{ \sin \left\{ \ln(x+1) \right\} \right\} \cos \left\{ \ln(x+1) \right\} \frac{1}{x+1} (1)$ $= \frac{-\cos ec \left\{ \sin \left\{ \ln(x+1) \right\} \right\} \cot \left\{ \sin \left\{ \ln(x+1) \right\} \right\} \cos \left\{ \ln(x+1) \right\}}{x+1}$	K1K1 J1

	For f to be continuous at $x = 0$, $f(0) = \lim_{x \rightarrow 0^-} e^{3x+d} = \lim_{x \rightarrow 0^+} 3x + 5$ $e^{3(0)+d} = 3(0) + 5$ $e^d = 5$ $d = \ln 5$	K1 K1 J1
bi	$f(x) = \begin{cases} x^2 - 1 & x \leq 1 \\ k(x-1) & x > 1 \end{cases}$ $f(1) = \lim_{x \rightarrow 1^-} x^2 - 1 = \lim_{x \rightarrow 1^+} k(x-1)$ $1^2 - 1 = k(1-1)$ $0 = 0$ thus, k can take any real value for the continuity, $k \in \mathbb{R}$	K1 K1 J1
bii	$f'(1^-) = \lim_{x \rightarrow 1^-} \frac{(x^2 - 1) - (1^2 - 1)}{x - 1}$ $= \lim_{x \rightarrow 1^-} \frac{x^2 - 1}{x - 1}$ $= \lim_{x \rightarrow 1^-} \frac{(x-1)(x+1)}{x-1}$ $= \lim_{x \rightarrow 1^-} (x+1)$ $= 1 + 1$ $= 2$ $f'(1^+) = \lim_{x \rightarrow 1^+} \frac{(k(x-1)) - (1^2 - 1)}{x - 1}$ $= \lim_{x \rightarrow 1^+} \frac{k(x-1)}{x-1}$ $= \lim_{x \rightarrow 1^+} k$ $= k$	K1 K1J1
	for f to be differentiable at $x=1$ $f'(1^-) = f'(1^+)$ $k = 2$	J1

11a	$f(x) = \frac{1}{x+1}$ $f'(x) = \lim_{h \rightarrow 0} \frac{\frac{1}{(x+h)+1} - \frac{1}{x+1}}{h}$ $= \lim_{h \rightarrow 0} \frac{x+1-x-h-1}{h((x+h)+1)(x+1)}$ $= \lim_{h \rightarrow 0} \frac{-h}{h((x+h)+1)(x+1)}$ $= \lim_{h \rightarrow 0} \frac{-1}{((x+h)+1)(x+1)}$ $= \frac{-1}{(x+1)(x+1)} = \frac{-1}{(x+1)^2}$	K1 K1 K1 J1
bi	$y \ln x = e^{x-y}$ $u = y \quad v = \ln x$ $u' = \frac{dy}{dx} \quad v' = \frac{1}{x}$ $\ln x \frac{dy}{dx} + y \left(\frac{1}{x} \right) = e^{x-y} \left(1 - \frac{dy}{dx} \right)$ $\ln x \frac{dy}{dx} + y \left(\frac{1}{x} \right) = e^{x-y} - e^{x-y} \frac{dy}{dx}$ $\ln x \frac{dy}{dx} + e^{x-y} \frac{dy}{dx} = e^{x-y} - \frac{y}{x}$ $\frac{dy}{dx} (\ln x + e^{x-y}) = e^{x-y} - \frac{y}{x}$ $\frac{dy}{dx} = \frac{e^{x-y} - \frac{y}{x}}{\ln x + e^{x-y}}$ $\frac{dy}{dx} = \frac{xe^{x-y} - y}{\ln x + e^{x-y}}$ $\frac{dy}{dx} = \frac{xe^{x-y} - y}{x(\ln x + e^{x-y})}$	K1 K1 J1

bii	$\frac{1}{y} - \frac{1}{x} = 3$ $y^{-1} - x^{-1} = 3$ $-y^{-2} \frac{dy}{dx} - (-1)x^{-2} = 0$ $-\frac{1}{y^2} \frac{dy}{dx} = -\frac{1}{x^2}$ $\frac{1}{y^2} \frac{dy}{dx} = \frac{1}{x^2}$ $\frac{dy}{dx} = \frac{y^2}{x^2}$ when $x = \frac{1}{2}$ $\frac{1}{y} - \frac{1}{\frac{1}{2}} = 3$ $\frac{1}{y} - 2 = 3$ $\frac{1}{y} = 5$ $y = \frac{1}{5}$ $\frac{dy}{dx} = \frac{\left(\frac{1}{5}\right)^2}{\left(\frac{1}{2}\right)^2} = \frac{4}{25}$	K1 K1 J1 J1
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12a	$x = \ln(1+t) \quad y = e^{t^2}$ $\frac{dx}{dt} = \frac{1}{1+t} \quad \frac{dy}{dt} = e^{t^2} (2t)$ $\frac{dt}{dx} = 1+t \quad \frac{dy}{dt} = 2te^{t^2}$ $\frac{dy}{dx} = 2te^{t^2} \times (1+t)$ $\frac{dy}{dx} = 2te^{t^2} (1+t)$ $u = 2te^{t^2} \quad v = 1+t$ $u = 2t \quad v = e^{t^2}$ $u' = 2 \quad v' = 2te^{t^2}$ $u' = 2e^{t^2} + 2t(2te^{t^2}) \quad v' = 1$ $\frac{d^2y}{dx^2} = \left((1+t)(2e^{t^2} + 2t(2te^{t^2})) + 2te^{t^2} (1) \right) \times (1+t)$ $\frac{d^2y}{dx^2} = (1+t)(2e^{t^2} + 4t^2e^{t^2} + 2te^{t^2} + 4t^3e^{t^2} + 2te^{t^2})$ $\frac{d^2y}{dx^2} = (1+t)(2e^{t^2} + 4t^2e^{t^2} + 4te^{t^2} + 4t^3e^{t^2})$ $\frac{d^2y}{dx^2} = 2e^{t^2} (1+t)(1 + 2t^2 + 2t + 2t^3)$	K1 K1 J1 J1 K1 J1
b	$\frac{dy}{dx} = 0$ $2te^{t^2} (1+t) = 0$ $t = 0 \quad t = -1$ since given that $t > -1$, the curve has only one extremum when $t = 0$ $t = 0$, $x = \ln(1+0) \quad y = e^{0^2}$ $x = 0 \quad y = 1$ $\therefore (0,1)$ is the extremum point	K1 J1 K1 J1

13a	$\left. \frac{d^2 y}{dx^2} \right _{t=0} = 2e^{0^2} (1+0) \left(1 + 2(0)^2 + 2(0) + 2(0)^3 \right) = 2 > 0$ (0,1) is minimum point $V = \pi r^2 h = 128\pi$ $h = \frac{128\pi}{\pi r^2}$ $h = \frac{128}{r^2}$ $S = 2\pi r^2 + 2\pi r h$ $S = 2\pi r^2 + 2\pi r \left(\frac{128}{r^2} \right)$ $S = 2\pi r^2 + 256\pi r^{-1}$ $\frac{dS}{dr} = 4\pi r - 256\pi r^{-2}$ $\frac{dS}{dr} = 0$ $4\pi r - 256\pi r^{-2} = 0$ $4\pi r - \frac{256\pi}{r^2} = 0$ $4\pi r = \frac{256\pi}{r^2}$ $r^3 = \frac{256\pi}{4\pi}$ $r = 4$ $h = \frac{128}{4^2} = 8$ $\frac{d^2 S}{dr^2} = 4\pi + 512\pi r^{-3}$ $\left. \frac{d^2 S}{dr^2} \right _{r=4} = 4\pi + 512\pi (4)^{-3} = 12\pi > 0$ h=8 and r=4 will give minimum surface are	K1 J1 K1 K1 K1 J1 K1 J1
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13b	$\frac{dV}{dt} = \frac{3}{20}$ $\tan 30^\circ = \frac{r}{h}$ $\frac{1}{\sqrt{3}} = \frac{r}{h}$ $r = \frac{h}{\sqrt{3}}$ $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$ $V = \frac{1}{3} \pi r^2 h$ $V = \frac{1}{3} \pi \left(\frac{h}{\sqrt{3}} \right)^2 h$ $= \pi \left(\frac{h}{\sqrt{3}} \right)^2 h$ $= \frac{\pi h^3}{9}$ $\frac{dV}{dh} = \frac{\pi h^2}{3}$ $\frac{dh}{dV} = \frac{3}{\pi h^2}$ $\frac{dh}{dt} = \frac{3}{\pi h^2} \times \frac{3}{20} = \frac{9}{20\pi h^2}$	K1 K1 K1 K1 K1 K1J1 J1